Vaibhav Singh

Machine Learning Engineer

Research → Train → Productionize

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Education

•Jawaharlal Nehru University, New Delhi ***CGPA: 6.8/9***

B.Tech - Electronics and Communications Engineering *2021 - 2025*

•GATE Data Science & AI Exam, 2025 ***All India Rank 398***

Experience

•REint Technologies Pvt Ltd ***Machine Learning Engineer***

08/2025 - Present

- Led research and development of new **trading strategies in the power market**, improving backtested **win rate from 51% to 57%** while maintaining drawdown under 13%
- Owned **evaluation of a client-facing LLM agent**, benchmarking proprietary and open-weight LLMs to **drive a production model switch** with 50% lower inference cost
- Researched and developed an **electricity price forecasting** system for the Indian power market, achieving **15% MAPE**; also built day-ahead and intraday **demand forecasting** systems achieving **3% MAPE**
- Improved client-facing **solar generation forecasting** for Ireland, cutting overall portfolio MAE by 11%

•Defense Research and Development Organization (Project) ***Generative AI***

Certificate of Experience *01/2025 - 05/2025*

- Collaborated on a **Joint Defence Research Project** between JNU and MoD focused on Remote Sensing and AI
- Delivered a GenAI solution to convert **Synthetic Aperture Radar (SAR)** signals to Optical images
- Improved all-weather object detection & reconnaissance through satellites using GAN based architecture

•Research Intern ***Non-Linearity in Deep Learning Models***

Certificate of Experience *06/2024 - 12/2024*

- Designed an **image-augmentation** technique to enhance non-linearity in CNNs, validating generalization gains through **ablation studies** across network depth, width, and activation functions
- Benchmarked against **reproduced SOTA baselines**, improving classification accuracy and segmentation **IoU by 1–2%** on CIFAR-10 and Aerial-View segmentation datasets

Personal Projects

•Research Bot (Generative AI) *06/2024 - 07/2024*

Github(Notebook) / Deployment

- Developed a **GenAI tool** to automate Literature Review generation with citations, reducing manual effort by 80%
- Integrated LLMs at multiple stages to retrieve and combine the latest papers using **LangChain**

•Chess AI (Deep RL) *08/2023 - 05/2024*

Github(Notebook) / Deployment

- Trained a Chess Bot that learns to play using **Deep Reinforcement Learning** and **achieved 1100 ELO Rating**
- Engineered a novel **highly compressed encoding** for model i/o, **reducing size by 49% and 85%** respectively
- Implemented a scalable **client-server architecture** capable of running 100+ parallel simulations

Technical Skills

Languages and Databases: Python, JavaScript, MySQL, DuckDB, PostgreSQL

Core ML: PyTorch, TensorFlow, scikit-learn, Computer Vision, GANs, Reinforcement Learning, Statistical Modeling, Data Analysis

LLMs & GenAI: Transformers, Hugging Face, LangChain, LangGraph, LLM Evaluation, Claude Code, Agentic Workflows

Web Development: Node.js, FastAPI, React.js

Deployment & MLOps: AWS EC2, Google Cloud Platform, Git, DVC, Docker, MLflow, ETL, Cloudflare Workers

Leadership Experiences and Achievements

•Vice President: Loop, The Developers' Club *09/2023 - 08/2024*

- Guided 20+ students through a **75-day Summer Mentorship Program** focused on AI/ML